GINWA FAKIH

AI & Knowledge Graph Engineer

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PROFILE

PhD in Computer Science specializing in data modeling, knowledge graphs and artificial intelligence. Expertise in ontologies, heterogeneous data integration and interoperability, as well as LLMs, RAG and GraphRAG. Proficient in RDF, OWL, SPARQL, SQL and Python, combining Data Modeling and generative AI.

EXPERIENCE

Semantic Web & Knowledge Graph Researcher (ATER) 09/2025 – 08/2026 | Nantes
LS2N Laboratory / Nantes Université

- PARTAGE project (CominLabs, ANR): contributed to an open-source platform for open educational resources, combining knowledge graphs, ontologies and LLMs for automatic metadata enrichment; developed APIs (FAIR principles, SPARQL, search engine).

Doctoral Researcher — Knowledge Graphs and Semantic Search 02/2023 – 09/2025 | Nantes
LS2N Laboratory

- Designed and deployed end-to-end large-scale data solutions based on knowledge graphs, ontologies and machine learning.
- Modeled metadata schemas, ontologies and knowledge graphs to improve data interoperability and governance.
- Designed and implemented retrieval, ranking and recommendation algorithms based on semantic similarity and graph embeddings.
- Developed graph data processing and evaluation pipelines independently, from design to validation.

Research Intern — Natural Language Processing 02/2021 – 07/2021 | Nantes
LS2N Laboratory

- Extracted and analyzed terms from text corpora to identify domain-specific topics.
- Used semi-supervised probabilistic modeling (SeededLDA) for topic discovery and classification.

PERSONAL PROJECTS

Industrial GraphRAG Copilot | 2026

Designed an RDF knowledge graph integrating simulated equipment, sensors, observations and maintenance data. Aligned identifiers, queried with SPARQL and combined structured and documentary data in a GraphRAG architecture. https://github.com/GhinwaFAKIH/industrial-graphrag-copilot

Research Paper Chatbot — RAG-based Document Assistant | 2026
Chatbot querying scientific papers in natural language, with answers grounded in source documents (RAG). Pipeline: PDF extraction, chunking, embeddings, FAISS, OpenAI/Llama generation, orchestrated with LangChain. github.com/GhinwaFAKIH/Research-Paper-Chatbot-RAG

SKILLS

Semantic Web: RDF, RDF-star, OWL, Linked Data, SPARQL, SHACL, Apache Jena, Virtuoso, RDF2Vec, RDFStar2Vec

Knowledge Graphs: ontology modeling, data interoperability and governance, data quality and validation, semantic similarity, graph embeddings

Generative AI & LLMs: RAG, GraphRAG, AI agents, LangChain, OpenAI API, Llama, prompt engineering, vector embeddings, FAISS, Sentence Transformers

Machine Learning: scikit-learn, TensorFlow, PyTorch, K-Means, NLTK, spaCy, clustering, classification

Programming: Python, Java, C/C++, SQL, SPARQL, TypeScript, HTML — JVM ecosystem

Data & Visualization: PostgreSQL, MySQL, GraphDB, Fuseki, Matplotlib, Seaborn, Plotly, Streamlit, PowerBI

Cloud & DevOps: AWS, GCP, Docker, Git, Linux

EDUCATION

PhD in Computer Science — Nantes Université 2022 – 2025 • Nantes
Semantic Web and Linked Data specialization — improving knowledge graph querying

MSc in Data Science — Polytech Nantes 2020 – 2021 • Nantes
Data mining, clustering and indexing, classification, probabilistic graphical models, visual analytics

BSc in Electrical, Electronics, Computer & Telecom Engineering — Lebanese University 2016 – 2020 • Beirut

LANGUAGES & OTHER

Languages: Arabic (native), English (fluent), French (B2) • Interests: travel, coding, photography, hiking

Certifications: Managing Machine Learning Projects with Google Cloud (Coursera) • French DELF — B2